

2023.

Q5.

$$(a). G_{ZAS}(z) = (1 - z^{-1}) \cdot z \left( \frac{G(s)}{s} \right)$$

$$\frac{G(s)}{s} = \frac{s-5}{s(3s+1)} = \frac{A}{s} + \frac{B}{3s+1} = \frac{(3A+B)s + A}{s(3s+1)}$$

$$\Rightarrow \begin{cases} 3A+B=1 \\ A=-5 \end{cases} \Rightarrow \begin{cases} A=-5 \\ B=16 \end{cases}$$

$$\begin{aligned} \text{Therefore, } z \left( \frac{G(s)}{s} \right) &= z \left( -5 \cdot \frac{1}{s} + \frac{16}{3} \cdot \frac{1}{s+\frac{1}{3}} \right) \\ &= -5 \cdot \frac{z}{z-1} + \frac{16}{3} \cdot \frac{z}{z-0.333} \\ &= \frac{z \cdot [-5(z-0.9355) + \frac{16}{3}(z-1)]}{(z-1)(z-0.9355)} \\ &= \frac{z}{z-1} \cdot \frac{\frac{1}{3}z - 0.6558}{z-0.9355} \\ &= \frac{1}{3} \cdot \frac{z}{z-1} \cdot \frac{z-1.9675}{z-0.9355} \\ G_{ZAS}(z) &= (1 - z^{-1}) \cdot z \left( \frac{G(s)}{s} \right) = 0.3333 \cdot \frac{z-1.9675}{z-0.9355} \end{aligned}$$

$$\begin{aligned} (b). C_L(z) &= \frac{R(z) \cdot G(z)}{1 + R(z) \cdot G(z)} = \frac{0.3333 \cdot \frac{z-1.9675}{z-1}}{1 + 0.3333 \cdot \frac{z-1.9675}{z-1}} \\ &= \frac{0.3333 \times (z-1.9675)}{z-1 + 0.3333z - 0.3333 \times 1.9675} \\ &= \frac{0.3333(z-1.9675)}{1.3333z - 1.6558} \\ &= 0.06249 \cdot \frac{z-1.9675}{z-0.3105} \end{aligned}$$

$$(c). C_L(z) = \frac{U(z)}{E(z)} = 0.06249 \cdot \frac{z-1.9675}{z-0.3105}$$

$$\Rightarrow U(z) = -0.3105 U(z) z^{-1} + 0.06249 E(z) - 0.06249 \times 1.9675 z^{-1}$$

$$\Rightarrow u_k - 0.3105 u_{k-1} = 0.06249 e_k - 0.1224 e_{k-1}$$

(d), When  $\omega T = 0.1$ , let  $z = e^{j\omega T}$

$$C_L e^{j\omega T} = 0.06249 \cdot \frac{e^{j\omega T} - 1.9675}{e^{j\omega T} - 0.3105} = 0.0883 \angle -2.894$$

$$\Rightarrow u_{ss} = 0.88 \sin(0.1k + 2.8945)$$

Q6.

(a).

$$(b). G_{cl}(z) = \frac{G(z)}{1+G(z)} = \frac{K}{z^2 + 0.9z + K}$$

$$\begin{cases} 0.9 = -2 \cos \omega_d T \cdot e^{-\zeta \omega_n T} \\ K = e^{-2\zeta \omega_n T} = 0.3679 \\ \zeta = \frac{1}{\zeta \omega_n} = 1 \end{cases}$$

$$(c). \omega_d = \frac{1}{T} \cos^{-1} \left( -\frac{0.9}{2} \cdot e^{\frac{T}{\zeta}} \right) = 4.8135 \text{ rad/s}$$

$$\omega_d = \omega_n \sqrt{1-\zeta^2} = \omega_n \cdot \sqrt{1 - \left(\frac{1}{\zeta \omega_n}\right)^2} \Rightarrow \omega_n = 4.9163 \text{ rad/s}$$

$$\zeta = \frac{1}{\zeta \omega_n} = 0.2034$$

$$\begin{aligned} (d). e_{ss} &= \lim_{z \rightarrow 1} (z-1) \left( \frac{K}{z^2 + 0.9z + K} \right) \frac{z}{z-1} \\ &= \frac{1.9}{1.9 + K} = \end{aligned}$$

Q1, (a) - ...

(b),

(i), Laplace transform:

$$JR s^2 \Theta(s) + k_t k_m s \Theta(s) = k_t V(s)$$
$$\Rightarrow G(s) = \frac{\Theta(s)}{V(s)} = \frac{k_t}{JR s^2 + (k_t k_m) s} = \frac{\frac{k_t}{JR}}{s^2 + \frac{k_t k_m}{JR} s}$$

(ii),

Observer canonical form,

$$A_o = \begin{bmatrix} -\frac{k_t k_m}{JR} & 1 \\ 0 & 0 \end{bmatrix} \quad B_o = \begin{bmatrix} 0 \\ \frac{k_t}{JR} \end{bmatrix}$$

$$C_o = [1 \ 0]$$

$$\begin{bmatrix} \theta' \\ \theta \end{bmatrix} = \dots$$

(iii) Any state that depends on initial conditions - ...

Q2, (a) Nyquist sampling theorem states that the sample rate  $f_s$  must be at least twice the bandwidth of the signal to

avoid aliasing.

Q3. (a). Nyquist stability criterion

uses the open loop frequency response to determine closed-loop stability by check whether the plot encircles the critical point  $-1$ .

(iv) 
$$V = \min_{\omega} |1 + SL(j\omega)| \quad \downarrow$$

still stable but ~~poorly~~ damped

with large overshoot and

high sensitivity to disturbance.